

PHASE 6 – PROJECT TESTING

Script-Controlled ACL – Testing and Verification of Record Access

1. Testing Overview

Testing is performed to verify whether the Script-Controlled ACL project correctly controls access to student records.

The testing checks the permissions of different users and verifies the Read, Write, Create, and Delete operations.

The main purpose is to ensure that authorized users can perform permitted operations and unauthorized users are restricted.

2. Testing Objectives

The main objectives of testing are:

- To verify user role-based access.
- To verify branch-based record access.
- To test Read access.
- To test Write access.
- To test Create access.
- To test Delete access.
- To verify administrator access.
- To identify unauthorized access attempts.
- To ensure that ACL scripts work according to the defined conditions.

3. Test Environment

The project is tested using the following environment:

Component	Details
Platform	ServiceNow
Access Control	ACL
Scripting	ACL Script
Table	Student Records
Branch Values	ECE, EEE, CSE
Operations Tested	Read, Write, Create, Delete

4. User Role Testing

Different users and roles are used during testing.

The role of each user is checked before accessing the Student Records table.

The testing verifies whether the user receives only the permissions allowed by the configured ACL conditions.

5. Branch-Based Access Testing

The project is tested using student records belonging to different branches.

For example:



An authorized EEE user is tested with EEE records and restricted records from other branches. This verifies that the branch value is correctly considered by the ACL script.

6. Read ACL Testing

The Read ACL is tested to verify whether users can view the permitted student records.

Test Process: Login with an authorized user → Open the Student Records table → Try to view permitted records → Try restricted records → Verify the access result.

Expected Result: Authorized records should be accessible, while restricted records should not be accessible.

7. Write ACL Testing

The Write ACL is tested to verify whether users can modify student records.

Test Process: Login with the required user → Open an accessible student record → Try to modify the record → Save the record → Verify whether the modification is allowed.

Expected Result: Only users satisfying the required ACL conditions should be allowed to modify records.

8. Create ACL Testing

The Create ACL is tested to verify whether users can create new student records.

Test Process: Login with the required user → Open the Student Records table → Select the option to create a new record → Enter the required information → Try to save the record → Verify the result.

Expected Result: Users with the required permission should be able to create records, while unauthorized users should be restricted.

9. Delete ACL Testing

The Delete ACL is tested to verify whether users can delete student records.

Test Process: Login with the required user → Open a student record → Try to delete the record → Verify the result.

Expected Result: Only users satisfying the required access conditions should be allowed to delete records.

10. Administrator Access Testing

Administrator access is also verified.

The administrator is tested with the Student Records table to confirm that the administrator has full access.

The following operations are checked:

- Read
- Write
- Create
- Delete

Expected Result: The administrator should have full access to the required student records and operations.

11. Test Case Summary

The following test cases are used to verify the configured access conditions:

Test Case	Operation	Condition Checked	Expected Result
TC01	Read	User role and branch	Access according to permission
TC02	Write	User role and branch	Modification allowed only when permitted
TC03	Create	User permission	Record creation controlled
TC04	Delete	User permission	Record deletion controlled
TC05	Branch Access	Branch value	Restricted branch access denied
TC06	Administrator	Administrator role	Full access

12. Testing Outcome

The testing process verifies the functionality of the Script-Controlled ACL system.

The project ensures that:

- Access is controlled using ACLs.
- User roles are checked.
- Branch values are considered for access.
- Read, Write, Create, and Delete operations are controlled.
- Unauthorized access is restricted.
- Administrator access is provided as required.

13. Conclusion

Testing confirms that the Script-Controlled ACL project provides controlled access to student records in ServiceNow. Different users, roles, branch values, and record operations are tested to verify that the configured ACL conditions work as expected.